Week 18: Hidden changes

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

The answers, proofs, held hints and audit table are in the print guide, week-18-facilitator.pdf, whose source is editable/facilitator-src/guide.json with route-note.json.

Problems and routes

Problem	K–1	2–3	4–5	route
Which keys always let the receiver find the picture?	P1	P1	P1	first
Which pictures could each final row mean?	P2			first
A row the receiver could get from either picture		P2		first
The shortest key for two pictures	P3	P3		return
Every three-counter key for two pictures	P4			return
Four-counter keys with two filled circles in each row		P4		return
A five-counter key for four pictures		P5	P2	2-3 return; 4-5 first
Every square row that works with a given triangle row	P5			return
The shortest key when two counters may change		P6		return
How many differing places keep two pictures apart?			P3	return
Every row the receiver might see from one row			P4	return
The shortest key for four pictures			P5	return
Can three pictures share a short key?	P6		P6	return

First route. K–1 P1–P2; Grades 2–3 P1–P2; Grades 4–5 P1–P2. **Return visits.** K–1 P3–P6; Grades 2–3 P3–P6; Grades 4–5 P3–P6.

Key, one entry per problem

Which keys always let the receiver find the picture?

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. Rows that differ in all three places (000 and 111) survive one hidden change; rows that differ in one place (001, 101), or two-place rows, do not.

Which pictures could each final row mean?

K–1 P2 · first route

Idea. With 000 and 111 every final row means one picture; with 00 and 11 the rows 01 and 10 could mean either.

A row the receiver could get from either picture

Grades 2–3 P2 · first route

Idea. Two rows that differ in one or two places share a possible received row; rows that differ in three or four places do not.

The shortest key for two pictures

K–1 P3 · Grades 2–3 P3 · return route · wording differs by band

Idea. Two pictures need rows of three counters: two-counter rows always share a possible received row.

Every three-counter key for two pictures

K–1 P4 · return route

Idea. The three-counter keys that always work are the eight ordered pairs of opposite rows.

Four-counter keys with two filled circles in each row

Grades 2–3 P4 · return route

Idea. Two rows with two filled circles each differ in an even number of places, so only opposite rows (four differences) work: six ordered keys.

A five-counter key for four pictures

Grades 2–3 P5 · Grades 4–5 P2 · 2-3 return; 4-5 first route · wording differs by band

Idea. Four rows of length five can differ pairwise in at least three places, for example 01001, 01110, 10000, 10111.

Every square row that works with a given triangle row

K–1 P5 · return route

Idea. A square row works exactly when it differs from the triangle row in at least three of the four places: the opposite row and its four one-change neighbours, five for every triangle row.

The shortest key when two counters may change

Grades 2–3 P6 · return route

Idea. With up to two hidden changes, two pictures need rows that differ in five places, so five counters.

How many differing places keep two pictures apart?

Grades 4–5 P3 · return route

Idea. Two rows can always be told apart after at most one change exactly when they differ in at least three places.

Every row the receiver might see from one row

Grades 4–5 P4 · return route

Idea. A row of length n can arrive as itself or with one of its n entries changed: n+1 received rows.

The shortest key for four pictures

Grades 4–5 P5 · return route

Idea. Four pictures need length five: four rows each with n+1 received rows need 4(n+1) at most 2 to the n, which fails for n up to 4.

Can three pictures share a short key?

K–1 P6 · Grades 4–5 P6 · return route · wording differs by band

Idea. Three pictures fit neither three counters nor four: three rows cannot differ pairwise in three places when each position adds at most two to the three distances.